

Hyperscale AI Data Center Proliferation in Municipal Watersheds: An Environmental, Technical, and Socioeconomic Assessment

**Case Focus: The City of Quinte West & Hastings County, Ontario
| Comparative Framework: Ontario (IESO), Quebec (Hydro-Québec),
Alberta (AESO), and United States (PJM/Virginia)**

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Executive Summary

This paper examines the infrastructure, ecological, and fiscal impacts of hyperscale and edge AI data center developments on mid-tier Canadian municipalities, centering on the City of Quinte West, Ontario. Rapid growth in AI training and high-density inference clusters has altered the utility demand profile of industrial facilities, escalating single-facility grid connection requests from traditional 5-10 megawatt (MW) baselines to continuous draws of 50-100+ MW. In Quinte West, municipal deliberations surrounding proposed conversions (specifically the 8 MW facility at 7 Riverside Drive, File D09/T09/26) highlight four systemic regulatory and infrastructure mismatches:

1. **Grid Allocation & Ratepayer Equity:** The continuous baseload of a single 8 MW facility matches the electricity consumption of approximately 7,500 single-family residences, placing sustained demand on local 230/44 kV distribution networks (Hydro One Sidney Transformer Station and Elexicon Energy) and accelerating provincial transmission reinforcement requirements.
2. **Hydrological & Chemical Vulnerability:** Although proposed closed-loop chiller systems avoid evaporative cooling losses (833,000 to 5.6 million L/MW/year)¹, bulk containment of propylene glycol heat-transfer fluids and on-site diesel generator fuel tanks poses severe Biochemical Oxygen Demand (BOD) risks to the Trent River and the Bay of Quinte watershed during containment failures.
3. **Acoustic and Public Health Externalities:** Standard municipal noise bylaws based on A-weighted decibels (dBA) fail to measure continuous low-frequency (<100 Hz) vibrational hum from rooftop chillers and substation transformers, which propagate through residential structures with minimal attenuation.
4. **The “Shell vs. Contents” Tax Deficit:** Under Section 3(1), paragraph 17 of the Ontario *Assessment Act*, all IT equipment, servers, and cooling machinery are legally exempt from municipal property assessment, meaning host cities derive tax revenue solely from the physical warehouse structure while servicing utility-scale industrial loads.

1 Local Power Grid Impacts & Transmission Constraints

1.1 Data Center Load Trajectories vs. Municipal Distribution Networks

Traditional industrial warehousing presents peak-to-average electricity load factors between 0.20 and 0.40. Hyperscale AI compute facilities run continuous non-linear baseload clusters with

¹Source: A moratorium on Ai data centres Canadians.pdf, Page 7

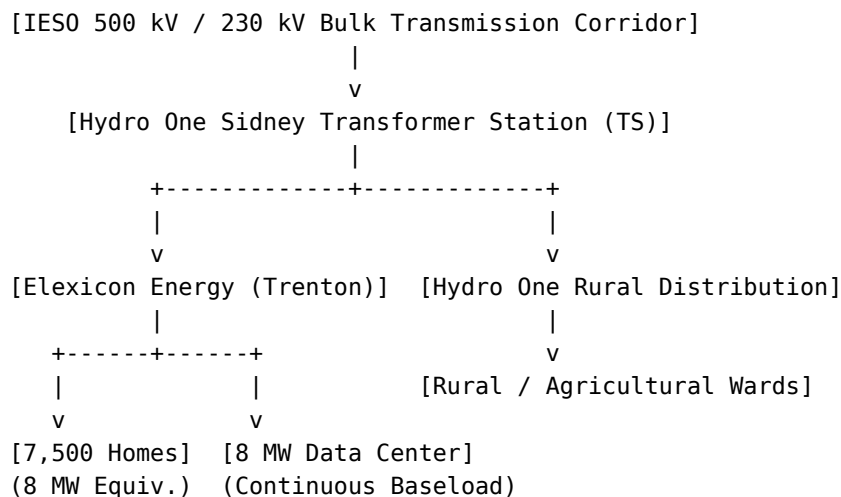
capacity factors exceeding 0.90 to 0.95. Across North America, the U.S. Department of Energy projects data centers will consume up to 12% of national electricity supply by 2028 (up from 4.4% in 2023, representing a 176 terawatt-hour increase), while facilities in Northern Virginia already command 26% of statewide electricity supply².

In Canada, uncapped interconnection requests have strained regional system operators. In Alberta, cumulative data center load requests surged to 20,000 MW against a total installed provincial generation capacity of only 10,000 MW, forcing the Alberta Electric System Operator (AESO) to institute a 1,200 MW supply cap to prevent grid instability³. In Ontario, the Independent Electricity System Operator (IESO) forecasts provincial electricity demand will expand by over 60% through 2050, driven by industrial electrification and high-density compute clusters.

1.2 Quinte West Power Inflow Architecture & Local Bottlenecks

The City of Quinte West receives bulk transmission power via Hydro One's 230 kV and 115 kV eastern transmission corridors (stretching between Cherrywood/Clarington Transformer Station and Kingston), stepping down through regional stations including the Sidney Transformer Station (TS). Local municipal retail distribution is divided geographically: Elexicon Energy distributes electricity to the urban Trenton core, while Hydro One Networks Inc. manages distribution across the surrounding wards (Frankford, Murray, and Sidney). Local run-of-the-river hydroelectric generation on the Trent River (including Frankford GS, Glen Miller GS, and Sidney GS) provides supplemental baseload generation but cannot absorb utility-scale industrial surges without bulk IESO grid imports.

The proposed 8 MW data center retrofit at 7 Riverside Drive requires a continuous allocation of 8,000 kW, matching the residential demand of approximately 7,500 homes in Hastings County. While existing 44 kV sub-transmission feeders can accommodate an initial 8 MW draw, expanding the campus or permitting additional facilities would exhaust local substation transformer capacity.



²Source: 5 ways data centers endanger their local communities and the country as a whole.pdf, Page 1

³Source: A moratorium on Ai data centres Canadians.pdf, Page 3

1.3 Transmission Expansion Costs & Ratepayer Cost-Shifting

When cumulative load exceeds substation headroom, bringing additional power into Quinte West requires major capital expenditure:

- **High-Voltage Line Construction:** Upgrading or constructing double-circuit 230 kV transmission lines in Ontario costs between \$2.5 million and \$4.5 million per kilometer.
- **Substation Reinforcement:** Adding a new 230/44 kV transformer station bay with gas-insulated switchgear costs between \$35 million and \$60 million.
- **Residential Ratepayer Exposure:** A 2023 infrastructure study estimated that utility and grid upgrades for data center clusters exceed \$150 million per 100 MW of capacity⁴. In Virginia, a 2024 legislative report found that data center grid expansion will increase typical residential electricity bills by \$14 to \$37 per month by 2040 (a 9% to 25% increase), disproportionately impacting lower-income households⁵. Nationally, Leger polling indicates 81% of Canadians oppose subsidizing data center utility costs through residential rate hikes⁶.

1.4 Canadian and International Legislative Frameworks

Governments and system operators have begun legislating strict interconnection rules:

- **Quebec (Hydro-Quebec):** Enacted interconnection tariffs requiring data center operators to curtail power consumption by up to 30% during peak winter heating events or pay non-firm industrial rates⁷.
- **Alberta (AESO):** Capped near-term large load interconnections at 1,200 MW while requiring upfront financial security from developers⁸.
- **Ontario (IESO & ERO Notice 025-1001):** Proposed revised connection cost recovery frameworks requiring large industrial loads to fund dedicated transmission reinforcements upfront via irrevocable letters of credit.
- **Federal Government (ISED):** Published *Canada's Responsible Data Centre Development Principles*, establishing national standards for grid additionality and community transparency.

2 Quinte West Nature, Watershed & Ecological Baselines

2.1 Major Waterways & Conservation Areas

The City of Quinte West sits at the confluence of major regional hydrological networks managed in coordination with Lower Trent Conservation (LTC):

- **The Trent River:** Serves as the southern discharge channel of the Trent-Severn Waterway, draining the Kawartha Lakes watershed directly through Frankford, Batawa, Glen Miller, and Trenton into the Bay of Quinte.

⁴Source: Health implications of the rapid rise of data centers in Virginia an exploratory assessment.pdf, Page 2

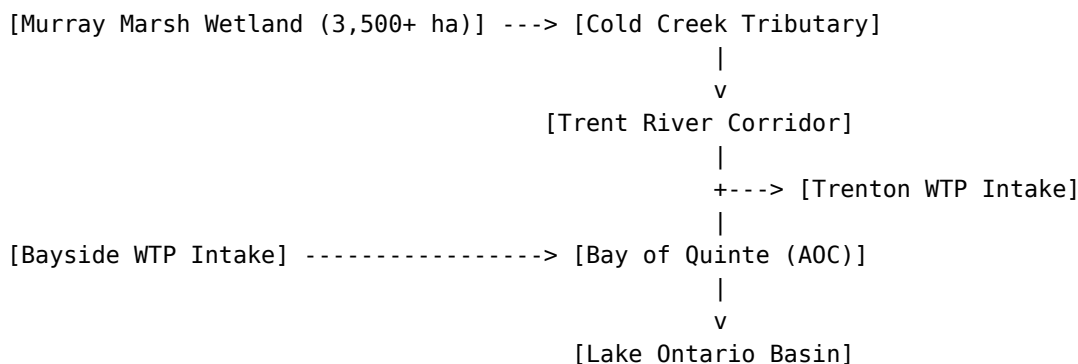
⁵Source: 5 ways data centers endanger their local communities and the country as a whole.pdf, Page 7

⁶Source: Leger Poll / Canadian Data Center Grid Demand

⁷Source: Canadian Climate Institute, Smart Way to Integrate AI Data Centres

⁸Source: A moratorium on Ai data centres Canadians.pdf, Page 3

- **The Bay of Quinte:** A Z-shaped embayment of Lake Ontario designated as a Great Lakes Area of Concern (AOC), characterized by shallow warm-water habitats sensitive to thermal plumes, nutrient loading, and oxygen depletion.
- **The Murray Canal:** An 8-kilometer excavated marine channel connecting the Bay of Quinte to Presqu'ile Bay on Lake Ontario, bordering sensitive coastal marshes.
- **Cold Creek & Local Tributaries:** Coldwater and coolwater streams feeding the Trent River system, supporting sensitive spawning habitats.
- **Lower Trent Conservation Areas:** Protected ecological reserves in Quinte West include the **Murray Marsh Natural Habitat Area** (over 3,500 hectares of regionally significant unfragmented wetland), **Sager Conservation Area**, and **Bleasdel Boulder Conservation Area**.



2.2 Municipal Drinking Water Sourcing & Infrastructure Baseline

Quinte West provides municipal potable water through two main surface water treatment systems:

- **Trenton Water Treatment Plant:** Draws raw surface water directly from the Trent River.
- **Bayside Water Treatment Plant:** Draws raw surface water from the Bay of Quinte.

Across its integrated municipal water and wastewater infrastructure, the City of Quinte West treats approximately **7,000 megalitres (ML) of drinking water annually** and **6,000 megalitres (ML) of wastewater annually**, maintaining **216 kilometres of water mains** and **8,800 metered service connections**⁹. Because the Trenton Water Treatment Plant intake sits downstream of industrial land along the Trent River corridor, industrial spills along Riverside Drive pose direct risks to municipal Intake Protection Zones (IPZ-1 and IPZ-2) under the Ontario *Clean Water Act*.

2.3 Wildlife Concentrations & Agricultural Apiary Corridors

- **Wildlife Concentrations:** Biodiversity is concentrated within the Murray Marsh wetland complex, the Lower Trent riparian corridor, and the coastal wetlands bordering the Murray Canal and Wellers Bay, providing critical habitat for migratory waterfowl, amphibians, and provincially tracked species at risk such as the Blanding's turtle (*Emydoidea blandingii*).
- **Apiaries & Pollinator Zones:** Commercial beekeeping (*Apis mellifera*) and pollination-dependent orchards are concentrated across the agricultural belts of **Murray Ward (Wooler**

⁹Source: City of Quinte West Water and Sewer FAQ

and Carrying Place corridors), Sidney Ward, and the Old Highway 2 / Bayside orchard corridor.

3 Environmental, Ecological & Water Table Impacts

3.1 Evaporative Water Cooling vs. Closed-Loop Cooling Systems

Data center thermal management relies on two primary cooling architectures:

- **Open-Loop Evaporative Cooling:** Dissipates server heat by evaporating water into the atmosphere. Hyperscale facilities using evaporative cooling consume **833,000 to 5.6 million litres of water annually per MW** of capacity¹⁰. In Virginia, data centers consumed over 2 billion gallons of water in 2023¹¹. Open-loop extraction lowers local water tables, reduces baseflow in adjacent streams, and concentrates dissolved solids in wastewater discharge.
- **Closed-Loop Chilled Water Systems:** Recirculate a sealed mixture of water and heat-transfer fluid (typically 30%-40% inhibited propylene glycol) through air-cooled rooftop chillers or dry coolers. Closed-loop systems eliminate continuous evaporative drawdown after initial charging, preserving the local water table. However, they require larger rooftop mechanical fan arrays (increasing low-frequency acoustic output) and maintain substantial volumes of chemical coolants on-site.

3.2 Surge Power Systems & Hydrocarbon Generator Emissions

To maintain continuous uptime during grid disturbances or IESO peak curtailments, data centers deploy multi-tiered backup systems combining Uninterruptible Power Supply (UPS) battery banks with banks of standby diesel generators:

- **EPA Tier 2 vs. Tier 4 Generator Emissions:** Operators routinely seek regulatory exemptions to install EPA Tier 2 diesel generators rather than Tier 4 Final units equipped with Selective Catalytic Reduction (SCR) and diesel particulate filters¹². Tier 2 diesel generators emit high volumes of nitrogen oxides (NOx), carbon monoxide (CO), unburned hydrocarbons, and fine particulate matter (PM2.5) during monthly load-bank testing and emergency operation¹³.

3.3 Quinte West Closed-Loop & Diesel Storage Spill Dynamics

For a closed-loop facility with diesel backup operating in Quinte West:

- **On-Site Storage Volumes:** An 8 MW facility requires approximately **40,000 to 75,000 litres of diesel fuel** (for 48-72 hours of full-load autonomy) and **15,000 to 30,000 litres of propylene glycol coolant**. A scaled 50-100 MW campus stores between **500,000 and 1,200,000+ litres of diesel fuel** and **150,000 to 350,000 litres of propylene glycol**.
- **Diesel Spill Impacts (Land & River):** A bulk fuel leak infiltrates permeable glacial till and alluvial soils, contaminating groundwater with benzene, toluene, ethylbenzene, and xylenes (BTEX) and polycyclic aromatic hydrocarbons (PAHs). Surface runoff reaching the Trent River

¹⁰Source: A moratorium on Ai data centres Canadians.pdf, Page 7

¹¹Source: 01_power_grid.md

¹²Source: 03_pollutants_spills.md

¹³Source: Diesel pollution from data centers.pdf

forms persistent hydrophobic surface films that coat shoreline vegetation, poison waterfowl, and force emergency shutdowns of downstream municipal drinking water intakes.

- **Propylene Glycol Spill Impacts (Aquatic Hypoxia):** Although propylene glycol has low acute mammalian toxicity, it poses a severe hazard to rivers and lakes due to its **extremely high Biochemical Oxygen Demand (BOD)**¹⁴. Microbial decomposition of propylene glycol consumes approximately **1.68 kg of dissolved oxygen per litre of glycol** released. If a closed-loop heat exchanger ruptures and drains into municipal storm sewers feeding the Trent River or Bay of Quinte, rapid bacterial oxygen consumption strips dissolved oxygen from the water column (dropping DO below 2.0 mg/L), triggering acute aquatic hypoxia and fish kills.

[Propylene Glycol Loop Rupture / Spill]

|
v

[Stormwater Discharge into Trent River / Bay of Quinte]

|
v

[Extreme Biochemical Oxygen Demand (BOD): 1.68 kg O₂ per Litre]

|
v

[Severe Dissolved Oxygen Depletion (<2.0 mg/L)] ---> [Fish Kills]

3.4 Risks to Local Wildlife and Honeybees

- **Pollinator & Honeybee Disruption:** Honeybees (*Apis mellifera*) rely on airborne and substrate-borne vibrations between **100 Hz and 500 Hz** (detected via Johnston's organ in their antennae) to interpret waggle dance foraging communications and coordinate colony defense¹⁵. Continuous low-frequency acoustic and ground-borne mechanical vibration from data center chillers and transformers masks these biological signals, increasing colony stress, reducing foraging efficiency, and impairing winter cluster thermoregulation.
- **Habitat Fragmentation, Light & Thermal Pollution:** 24/7 perimeter security floodlighting disrupts nocturnal insect navigation and migratory bird flight paths, while large impervious roof and pavement footprints create localized thermal heat islands.

4 Acoustic Concerns: dBA vs. dBC, Mitigation & Property Values

4.1 dBA vs. dBC Weighting Curves Applied to Data Centers

Sound level measurements use frequency weighting filters that alter how low-frequency acoustic energy is recorded:

- **A-Weighted Decibels (dBA):** Designed to approximate human hearing response to quiet sounds, the dBA filter aggressively subtracts energy below 1,000 Hz—attenuating sound at 63 Hz by -26.2 dB and at 31.5 Hz by -39.4 dB.

¹⁴Source: Vermont DEC Propylene Glycol Environmental Fact Sheet

¹⁵Source: NCBI PMC8708298

- **C-Weighted Decibels (dBC):** Maintains a nearly flat frequency response across the low-frequency spectrum (-0.8 dB at 63 Hz and -3.0 dB at 31.5 Hz), capturing the true physical sound pressure generated by heavy industrial fans, compressors, and transformers.

Frequency Band	dBA Filter	dBC Filter	Low-Frequency Gap (dBC - dBA)
31.5 Hz	-39.4 dB	-3.0 dB	+36.4 dB
63 Hz	-26.2 dB	-0.8 dB	+25.4 dB
125 Hz	-16.1 dB	-0.2 dB	+15.9 dB
250 Hz	-8.6 dB	0.0 dB	+8.6 dB
1,000 Hz	0.0 dB	0.0 dB	0.0 dB

Because closed-loop dry coolers and rooftop chillers emit concentrated acoustic energy between 31.5 Hz and 125 Hz, a data center measuring a legally compliant **45 dBA** at a residential lot line can simultaneously generate **65 to 72 dBC** of low-frequency hum¹⁶.

4.2 Data Center dBC Sources & Engineering Mitigations

- **Primary Sources of dBC:** Rooftop air-cooled chiller screw/scroll compressors, large-diameter axial condenser fans (blade-pass frequency hum), step-down substation transformers (120 Hz magnetostriction hum), and diesel generator exhaust stacks.
- **Mitigation Engineering:** Because low-frequency sound waves (3 to 11 metres in wavelength) diffract over standard wooden fences and penetrate residential framing, mitigation requires **mass-loaded masonry acoustic enclosures, deep-baffle dissipative silencers, spring-isolated equipment mounts, variable-frequency drives (VFDs) tuned to avoid harmonic beat frequencies, and minimum 300-to-500-metre setbacks** from sensitive receptors.

4.3 Medical Implications, Property Values & Noise Legislation

- **Medical Implications of dBC Exposure:** Low-frequency noise penetrates exterior walls and induces standing-wave resonance inside bedrooms. Short-term exposure causes ear pressure, headache, tinnitus, and sleep fragmentation; chronic long-term exposure triggers nocturnal cortisol elevation, autonomic nervous system dysregulation, and increased risk of hypertension and cardiovascular disease¹⁷.
- **Impact on Property Values:** Real estate analyses in Northern Virginia and Chandler, Arizona document residential property value reductions of **8% to 18%** for homes located within 400 metres of unmitigated data center chiller yards, accompanied by extended listing durations.
- **Acoustic Legislation:** Ontario's provincial noise guideline (**NPC-300**) and Quinte West By-law 12-030 rely primarily on dBA limits (typically 45 dBA nighttime / 50 dBA daytime), leaving

¹⁶Source: 04_acoustics_dba_dbc.md

¹⁷Source: How noise pollution quietly affects your health.pdf

a regulatory blind spot for low-frequency dBC emissions unless municipalities adopt supplemental C-weighted caps (such as limiting dBC - dBA differentials to ≤ 10 -15 dB).

5 Human Health Risks (Physical & Mental Epidemiology)

5.1 Physical Health Impacts

Data center operations affect human physical health through two primary pathways:

1. **Airborne Particulate & Photochemical Pollution:** Diesel backup generator testing and increased fossil-fuel grid dispatch release fine particulate matter (PM_{2.5}), nitrogen oxides (NO_x), and volatile organic compounds (VOCs)¹⁸. PM_{2.5} crosses the alveolar-capillary barrier into the systemic circulation, increasing rates of pediatric and adult asthma, chronic bronchitis, myocardial infarction, and stroke¹⁹.
2. **Cardiovascular & Autonomic Strain:** Chronic exposure to 24/7 environmental noise elevates systolic and diastolic blood pressure and promotes vascular inflammation²⁰.

5.2 Mental Health Impacts

Continuous non-stop tonal hum and low-frequency vibration induce documented psychological and neurological effects:

- **Chronic Noise Annoyance & Sleep Deprivation:** Inability to escape low-frequency hum inside one's home causes chronic insomnia, daytime cognitive fatigue, irritability, and clinical anxiety²¹.
- **Loss of Restorative Environment:** Persistent industrial noise erodes residential quality of life and triggers chronic stress responses across adjacent neighborhoods.

5.3 Health-Focused Regulatory Responses

In Washington State, the Department of Ecology mandated formal Health Risk Assessments (HRAs) for data center diesel emissions to protect downwind communities from carcinogenic diesel particulate matter²². The World Health Organization (WHO) *Environmental Noise Guidelines* explicitly note that low-frequency noise requires stricter nighttime thresholds than standard dBA metrics provide.

6 Municipal Business Tax Revenue (Shell vs. Contents)

6.1 Ontario MPAC Assessment Rules: The Machinery Exemption

In Ontario, property tax assessments for Quinte West are conducted by the Municipal Property Assessment Corporation (MPAC) under the **Ontario Assessment Act (R.S.O. 1990, c. A.31)**.

¹⁸Source: Analyzing air pollution health economic risks from AI data centers Harvard TH Chan School of Public.pdf

¹⁹Source: Data centers air pollution associated with lung issues Report.pdf

²⁰Source: 05_human_health.md

²¹Source: How Are Data Centers Impacting Health uconn.pdf

²²Source: 03_pollutants_spills.md

- **Real Property vs. Personal/Machinery Property:** Under **Section 3(1), paragraph 17 of the Ontario Assessment Act**, “*all machinery and equipment used for manufacturing or producing purposes*” is statutorily exempt from municipal property taxation²³.
- **Application to Data Centers:** MPAC assesses only the **land and the physical building shell** (concrete slab, structural steel frame, exterior walls, basic lighting, and standard building HVAC). The high-value interior contents—including AI GPU clusters, server racks, network switches, uninterruptible power supplies (UPS), and specialized process cooling skids—are classified as exempt machinery/personal property and taxed at **\$0** municipally.

Total Advertised Data Center Capital Investment (\$300M - \$1B+)

```
|
+---> [Contents / Personal Property: 80% to 90% of Value]
|   - AI Servers, GPUs, Networking Racks
|   - Process Chiller Skids & UPS Arrays
|   +---> MUNICIPAL TAX TO QUINTE WEST: $0 (Exempt under Assessment Act)
|
+---> [Building Shell & Land: 10% to 20% of Value]
|   - Industrial Acreage, Concrete Slab, Steel Warehouse Shell
|   +---> MUNICIPAL TAX TO QUINTE WEST: Standard Industrial Mill Rate
```

6.2 Comparative Global Tax Approaches

- **Canadian Municipalities:** Because Canadian provinces do not levy a municipal “Business Personal Property Tax” on interior computer hardware, a data center generates roughly the same property tax revenue per square foot as an unconditioned storage warehouse, while consuming up to 100 times the electrical and emergency infrastructure capacity.
- **United States Jurisdictions:** Many U.S. states (such as Virginia) allow counties to levy a local **Business Personal Property Tax** directly on the assessed value of the servers and cooling equipment inside the building (subject to rapid depreciation schedules), though states simultaneously lose hundreds of millions of dollars annually through state sales-and-use tax exemptions on server purchases²⁴.

7 Employment & Possible Spin-Off Industries

7.1 Headcount per Facility Size (Construction vs. Permanent Operations)

Data centers are among the most capital-intensive, lowest-labor-density industrial land uses:

- **Empirical Staffing Ratios:** An analysis of 1,200 U.S. data center facilities found that even massive hyperscale campuses employ fewer than 150 permanent workers, while standard regional facilities employ **5 to 30 permanent staff** (roughly **0.5 to 2.0 permanent employees per MW**)²⁵.
- **Quinte West Project Baseline:** For the proposed 8 MW facility at 7 Riverside Drive in Trenton, planning submissions indicated a total permanent workforce of **14 to 16 full-time employees**.

²³Source: MPAC Standard Industrial Properties / Ontario Assessment Act

²⁴Source: From Energy Use to Air Quality the Many Ways Data Centers Affect US Communities.pdf, Page 2

²⁵Source: From Energy Use to Air Quality the Many Ways Data Centers Affect US Communities.pdf, Page 9

- **Subsidy-to-Job Deficit:** While development creates 200-300 temporary construction jobs over 12-24 months, public infrastructure and tax subsidy costs average **\$50,000 to \$1,000,000+ per permanent job created**²⁶.

7.2 Construction Trades, Operational Job Titles & Local Procurement

- **Construction & Commissioning Trades:** High-voltage electricians (IBEW), commercial pipefitters and steamfitters, refrigeration and HVAC mechanics (313A license), structural iron-workers, millwrights, concrete finishers, and fiber-optic splicing technicians.
- **Long-Term Operational Job Titles:** Critical Facilities Technician (CFT), Data Center Operations Manager (\$74,000-\$160,000 salary range)²⁷, HVAC/Chiller Maintenance Engineer, Network Technician, and Contract Physical Security Officer.
- **Local Business Services Required:** Contracted 24/7 physical security, commercial snow removal and landscaping, licensed electrical thermography/transformer testing, fire suppression inspection, and hazardous waste hauling.
- **Consumable Materials Purchased:** Inhibited propylene glycol heat-transfer fluid, MERV-13/HEPA air handling filters, ultra-low-sulfur diesel (ULSD) fuel, lead-acid or lithium-ion replacement battery modules, dielectric transformer oils, and replacement server optics (procured almost exclusively from global OEMs rather than local vendors)²⁸.

8 Emergency Response & Secondary Municipal Hazards

8.1 Spill & Hazmat Response Agencies Beyond 911 in Quinte West

When a chemical, fuel, or battery hazmat emergency occurs in Quinte West, response coordination extends beyond Quinte West Fire & Emergency Services, OPP, and Hastings-Quinte Paramedic Services to specialized statutory agencies:

1. **Ontario Ministry of the Environment, Conservation and Parks (MECP) — Spills Action Centre (SAC):** Mandatory 24/7 provincial reporting authority (1-800-268-6060) under Part X of the *Environmental Protection Act* for all chemical, coolant, and hydrocarbon spills²⁹.
2. **Lower Trent Conservation (LTC):** Coordinates watershed flood and spill modeling and issues Drinking Water Source Protection alerts to shut raw water intakes at the Trenton and Bayside Water Treatment Plants.
3. **CANUTEC (Canadian Transport Emergency Centre):** Federal Transport Canada emergency technical center providing chemical plume, reactivity, and suppression guidance to municipal fire commanders³⁰.
4. **Hastings Prince Edward Public Health (HPEPH):** Issues public drinking water advisories and shelter-in-place orders during toxic smoke or chemical plume events.

²⁶Source: From Energy Use to Air Quality the Many Ways Data Centers Affect US Communities.pdf, Page 12

²⁷Source: From Energy Use to Air Quality the Many Ways Data Centers Affect US Communities.pdf, Page 9

²⁸Source: From Energy Use to Air Quality the Many Ways Data Centers Affect US Communities.pdf, Page 9

²⁹Source: Ontario Spills Action Centre MECP

³⁰Source: CANUTEC Emergency Response

5. **Environment and Climate Change Canada (ECCC) – National Environmental Emergencies Centre:** Enforces Section 36 of the federal *Fisheries Act* prohibiting the deposit of deleterious substances into fish-bearing waters (Trent River and Bay of Quinte).

8.2 Contamination Vectors & BESS Thermal Runaway Precedents

- **Contamination Types:** Data centers pose four major contamination risks: (1) bulk diesel fuel hydrocarbons (BTEX/PAHs), (2) propylene glycol coolant runoff, (3) per- and polyfluoroalkyl substances (PFAS) from fluorinated refrigerants and firefighting foams, and (4) **Battery Energy Storage System (BESS) thermal runaway byproducts**³¹.
- **BESS Thermal Runaway & Toxic HF Gas:** Lithium-ion battery banks undergoing thermal runaway vent explosive hydrogen gas, carbon monoxide, and **highly toxic hydrogen fluoride (HF) gas**, while contaminating firefighting water runoff with heavy metals (cobalt, nickel, manganese) and hydrofluoric acid³².
- **Canadian & International Accident Precedents:** Notable incidents include repeated diesel fuel leaks at hyperscale clusters in Northern Virginia (Loudoun County), coolant and sulphuric acid releases at U.S. and European facilities, and utility-scale lithium-ion battery fires in Arizona (McMicken), California (Moss Landing), and Europe that forced multi-day community evacuations and revealed municipal fire department equipment shortfalls.

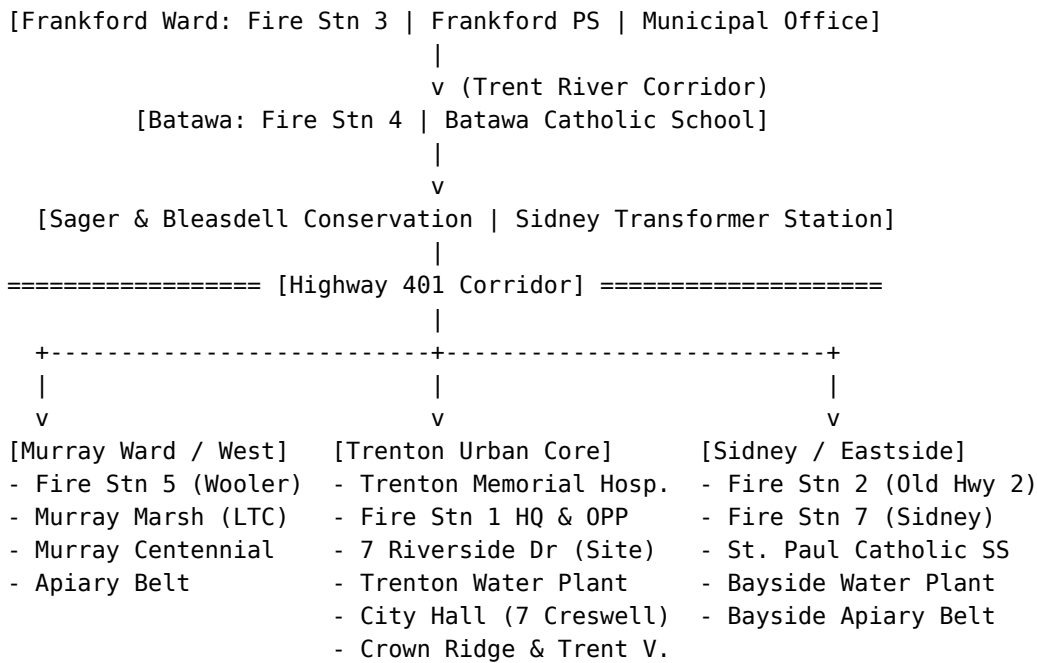
9 Quinte West Sensitive Receptors & Community Infrastructure Inventory

The following table and schematic map the locations of medical, educational, retirement, emergency, and municipal government facilities across Quinte West relative to industrial corridors:

³¹Source: 08_hazards_emergency.md

³²Source: 08_hazards_emergency.md

Category	Facility Name	Civic Address / Ward	Operational Role & Sensitivity
Medical	Trenton Memorial Hospital	242 King St, Trenton	Acute care & ER; diesel PM2.5 & power sensitivity
Medical	Trenton Community Health Centre	69 Catherine St, Trenton	Primary & community care clinic
Retirement	Crown Ridge Place LTC	106 Crown St, Trenton	Senior care; high sensitivity to nighttime dBC hum
Retirement	Trent Valley Lodge LTC	195 Bay St, Trenton	Waterfront senior care along Trent River mouth
Retirement	Seasons Dufferin Centre	344 Dufferin Ave, Trenton	Retirement residence
Education	Trenton High School (HPEDSB)	15 Fourth Ave, Trenton	Secondary school receptor
Education	St. Paul Catholic SS	1515 Whites Rd, Trenton	Secondary school receptor
Education	Prince Charles & Marc Garneau PS	Trenton Urban Core	Elementary school receptors
Education	Frankford / Batawa / Murray PS	Ward Corridors	Ward elementary receptors
Fire Dept	Quinte West Fire Station 1 (HQ)	49 Dixon Dr, Trenton	Career/composite hazmat & fire headquarters
Fire Dept	Quinte West Fire Station 2	165 Old Hwy 2, Trenton	Eastside response station
Fire Dept	Quinte West Stations 3-7	Wards 2, 3, & 4	Composite/volunteer ward stations
Police	Quinte West OPP Detachment	30 Dixon Dr, Trenton	Municipal & highway policing headquarters
Gov Office	Quinte West City Hall	7 Creswell Dr, Trenton	Municipal administration & Council chambers
Gov Office	Lower Trent Conservation	714 Murray St, Trenton	Watershed & source water protection authority



10 Actionable Policy Recommendations for Quinte West City Council

To protect municipal infrastructure, watershed integrity, and residential ratepayers, Quinte West City Council should enact five binding regulatory requirements prior to approving data center zoning or site-plan applications:

1. **Adopt Dual dBA/dBC Municipal Noise Limits:** Amend Quinte West Noise By-law 12-030 to enforce a nighttime residential boundary cap of **45 dBA and 55 dBC** (or mandate that C-weighted minus A-weighted measurements not exceed 10 dB), requiring pre-commissioning 3D acoustic modeling and continuous fenceline sound monitoring.
2. **Mandate Closed-Loop Cooling with 110% Secondary Glycol Containment:** Prohibit evaporative potable water cooling and require impermeable secondary containment vaults under all chiller loops and fuel tanks, sized to 110% of total fluid volume with automatic stormwater shutoff valves to protect the Trent River and Bay of Quinte.
3. **Require EPA Tier 4 Final Backup Generators:** Prohibit Tier 2 standby diesel exemptions and restrict routine generator load testing to daytime weekday hours under dispersion-favorable Air Quality Health Index (AQHI) conditions.
4. **Enforce Upfront Utility Cost-Recovery & Decommissioning Bonds:** Require developers to post irrevocable letters of credit covering 100% of Hydro One and Elexicon substation upgrades and site decommissioning costs.
5. **Institute a BESS Hazmat Readiness Levy:** Require facilities deploying lithium-ion BESS arrays to fund specialized gas-monitoring, thermal imaging, and containment equipment for Quinte West Fire & Emergency Services.

10.1 Comparative Case Study: Cambridge TOR1 vs. Secaucus Equinix NY2

To understand how facility failures threaten municipal watersheds, table 8.1 contrasts the July 2026 Cambridge Ascent TOR1 stormwater discharge with the September 2026 Secaucus Equinix NY2 bulk diesel release:

Incident Dimension	Cambridge Ascent TOR1 (Ontario, Canada – July 2026)	Secaucus Equinix NY2 (New Jersey, USA – September 2026)
Primary Contaminant	Suspected closed-loop chiller fluid / biological flush	No. 2 ultra-low-sulfur diesel fuel (~5,000–5,500 gallons)
Mechanism of Failure	Direct retention pond discharge / unmonitored storm outfall	Broken transfer valves and automated fuel system malfunction
Environmental Threat	Extreme Biochemical Oxygen Demand (BOD) & acute aquatic hypoxia	Hydrocarbon sheen, wetland degradation, PAH soil adsorption
Drinking Water Impact	Downstream travel modeling for Brantford municipal surface intake	Estuarine tidal dispersion into Hackensack River and Newark Bay
Regulatory Jurisdiction	Ontario MECP Spills Action Centre & Grand River Cons. Authority	NJ DEP, Secaucus Fire Hazmat & Clean Harbors emergency response
Municipal Lesson	Stormwater ponds are ineffective primary chemical barriers	Secondary containment basins must feature positive shutoff valves